Alex

Do not write in this margin

Q1

(a) 9, 15, 5, 11, 6

(b)

D	2	2	2
1	6	8	8
2	5	13	13
3	8	21	-1

OUTPUT: ~~6, 8~~ 6, 8

(c) run time error. ~~The~~ There will not be output as nums[5] isn't in the range of the array nums.

(d) He should use nums[i] to double the each valid element in the array.

Q2

(a)

```
~~RETURNS INTEGER STRING~~
~~FUNCTION CountEven (ARRAY Data) Data : ARRAY~~
    ~~DECLARE Count : INTEGER DECLARE Result : STRING~~
    ~~Count ← 2 Result ← ""~~
    WHILE ~~Data(Count)~~ ← LENGTH(Data)
        ~~OUTPUT Data(Count)~~ ~~Result ← Result & "," & Data[Count]~~
        ~~Count ← Count + 2~~
    ~~ENDWHILE~~
    ~~RETURN Result~~
~~ENDFUNCTION~~
```

(b)

```
FUNCTION CountEven RETURNS INTEGER
    DECLARE Count, i : INTEGER  Count ← 0  i ← 1
    WHILE ~~Data[Count]~~ i <= LENGTH(Data)
        IF Data[~~Count~~ i] MOD 2 = 0 THEN
            Count ← Count + 1
        i ← i + 1
        ENDIF
    ENDWHILE
    RETURN Count
ENDFUNCTION
```

write
this
margi

DECLARE Temp

Q2(b) ~~T1, T2, T3, T4, T5, T6, T7, T8, i : INTEGER~~

```
DECLARE Avr : REAL          DECLARE T : ARRAY[1:8] OF INTEGER
                                  Avr ← 0
For i ← 1 to 8
    INPUT Temp
    T[i] ← Temp
Next i
CALL DoubleAverage (T : ARRAY[1:8] OF INTEGER)
Avr ← DoubleAverage (T : ARRAY[1:8] OF INTEGER), OUTPUT Avr
FOR i ← 1 to 8
    IF Avr < T[i] THEN
        OUTPUT T[i] & " "
    ENDIF
NEXT i
```

Q3B (b) BYREF represents that the address of the variable Data is the same in the procedure and main. Hence if the programmer adjust Data in the procedure, Data will also be changed in main. The output will be "3 8 10" if BYREF is removed because A will not be adjusted by the procedure.

(a) A = [6, 5, 14, 9, 11, 10]

OUTPUT = 6 9 11

(c)
```
PROCEDURE Replace (BYREF Data : ARRAY[1:8] OF INTEGER,
OldVal : INTEGER, NewVal : INTEGER)
    DECLARE i : INTEGER
    FOR i ← 1 to 8
        IF Data[i] = OldVal THEN
            Data[i] ← NewVal
        ENDIF
    NEXT i
ENDProcedure
```

Q3B PROCEDURE

(d)
```
Swap(BYREF X = INTEGER, BYREF Y = INTEGER)
    DECLARE Container : INTEGER
    Container ← X
    X ← Y
    Y ← Container
ENDPROCEDURE
A ← 5
B ← 9
CALL Swap(A, B)
OUTPUT A, " ", B
```

OUTPUT =  9 5